BITS PILANI, DUBAI CAMPUS
SECOND SEMESTER 2025 – 2026
Department of Computer Science

Course Code: CS F429 | **Max Marks: 40**
Course Title: Natural Language Processing | **Weightage: 40%**
Mode of Assessment: Open Book | **Nature: Team-Based**

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NLP Research Assignment

Research Pipeline on Hallucination in Retrieval-Augmented Language Models (RALMs)

This assignment requires each team to conduct an original research study on hallucination in Retrieval-Augmented Language Models (RALMs), including formal problem definition, implementation, rigorous evaluation, and reproducible documentation.

1. **Team Structure**
   – Team size: 3–4 students
   – Each team must select ONE research track (Track A or Track B)
   – Equal distribution across both tracks will be maintained
   – Topic selection is first-come, first-approved

2. **Research Tracks**
   Each team must select **one** of the following research tracks and complete all specified requirements. Deviation from the defined objectives, datasets, and evaluation protocol is not permitted.

2.1. Track A: Dynamic Uncertainty-Aware Attribution
Quantifying Hallucination via Token-Level Information Gain in Retrieval-Augmented Language Models

This track focuses on designing a token-level uncertainty-based metric to quantify hallucination in RALMs. The central research question is:

Can changes in predictive uncertainty induced by retrieved context serve as a reliable signal of faithfulness or hallucination?

Teams are expected to formulate hallucination detection as an uncertainty quantification problem in the output probability space.

2.2. Required Technical Components (Track A)

Each team must:

1. Develop a token-level metric capturing the effect of retrieval on model uncertainty.
2. Conduct temporal analysis across generated sequences to study patterns preceding hallucinated spans.
3. Perform statistical validation against ground-truth hallucination labels.

4. Evaluate robustness across at least two datasets or domains.
5. Compare against at least two baselines (e.g., entropy-only and one external hallucination detection method).

Superficial entropy comparisons without deeper analysis will not receive full credit.

2.3. Track B: Pre-Generation Causal Drift Metric

Mechanistic Attribution for Faithfulness in Retrieval-Augmented Language Models

This track focuses on developing an internal representation-based metric that detects semantic drift prior to hallucinated outputs. The central research question is:

Do hidden-state dynamics within transformer layers provide predictive signals of hallucination before it manifests in generated text?

Teams must investigate internal representation shifts using principled attribution and intervention methods.

2.4. Required Technical Components (Track B)

Each team must:

1. Define a representation-based drift metric operating in hidden-state space.
2. Identify critical layers or components contributing to semantic drift.
3. Conduct causal intervention experiments (e.g., activation steering, suppression, or controlled ablation).
4. Demonstrate predictive power prior to hallucinated tokens.
5. Validate stability across at least two domains or datasets.

Attention heatmaps alone are insufficient. Quantitative and causal evidence is required.

2.5. Common Research Requirements (Applicable to All Tracks)

All teams must satisfy the following theoretical standards:

1. **Formal Metric Definition** Clearly define the proposed metric using precise mathematical notation.
2. **Specification of Components**
   Explicitly describe:
   - Input distributions or representations
   - The transformation or comparison function
   - The output quantity being measured
3. **Theoretical Justification**: Provide a principled explanation for why the proposed metric captures hallucination or semantic drift.
4. **Testable Hypothesis**: Clearly state a hypothesis that can be empirically validated (e.g., predictive power prior to hallucinated spans).
5. **Statistical Validation**: Report appropriate statistical measures, including confidence intervals or significance testing where applicable.

Superficial implementation without formal definition, theoretical grounding, and empirical validation will not receive full credit.

3. **Required Datasets (Both Tracks)**
   Teams must use at least two of the following:
   - RAGTruth
   - HaluEval
   - FEVER (optional extension)

   Use of additional datasets requires prior approval.

4. **Required Reading:**

Track A: Dynamic Uncertainty-Aware Attribution (Information Gain in RAG)

[1] Lewis et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. NeurIPS.
[2] Guu et al. (2020). REALM: Retrieval-Augmented Language Model Pre-Training. ICML.
[3] Izacard et al. (2022). Atlas: Few-shot Learning with Retrieval Augmented Language Models. ICML.
[4] Manakul et al. (2023). SelfCheckGPT: Zero-Resource Black-Box Hallucination Detection for Generative LLMs. EMNLP.
[5] Li et al. (2023). HaluEval: A Large-Scale Hallucination Evaluation Benchmark for LLMs. EMNLP.
[6] Xu et al. (2024). RAGTruth: A Hallucination Corpus for Trustworthy Retrieval-Augmented LMs. ACL Findings.
[7] Kadavath et al. (2022). Language Models (Mostly) Know What They Know. ACL.
[8] Guo et al. (2017). On Calibration of Modern Neural Networks. ICML.
[9] Jiang et al. (2022). Selective Prediction in NLP. ACL.
[10] Desai & Durrett (2021). Confidence Estimation for Neural Sequence Models. ACL.
[11] Tian et al. (2023). Trustworthy LLMs: Confidence and Calibration in Language Models. ACL.
[12] Malinin & Gales (2023). Uncertainty Estimation in Text Generation: A Survey. TACL.
[13] Dai et al. (2026). Reinforcement Learning with Fine-grained Knowledge Verification. AAAI (Feb 2026).
[14] Shelmanov et al. (2025). LM-Polygraph: Uncertainty Quantification for Large Language Models. ACL Tutorial (July 2025).
[15] Vectara (2025). FaithJudge: Benchmarking LLM Faithfulness in RAG with Evolving Human Annotations. EMNLP Industry (Nov 2025).
[16] Chen et al. (2025). Can LLMs Evaluate Complex Attribution in QA? A New Benchmark. ACL (Aug 2025).

Track B: Pre-Generation Causal Drift Metric (Mechanistic RAG)

[1] Anthropic (2021). A Mathematical Framework for Transformer Circuits.
[2] Meng et al. (2023). Causal Tracing for Language Models. ICLR.
[3] Nanda et al. (2022). Logit Lens: Interpreting Hidden States in Transformers. NeurIPS Workshop.
[4] Geva et al. (2021). Transformer Feed-Forward Layers Are Key-Value Memories. EMNLP.
[5] Jain & Wallace (2019). Attention is Not Explanation. NAACL.
[6] Turner et al. (2024). Activation Steering in Large Language Models. ICLR.
[7] Elhage et al. (2023). Interpretability in the Wild: A Circuit Analysis of GPT-2. ICLR.
[8] Wang et al. (2023). Attribution Methods for Generative Models. EMNLP.
[9] Zhao et al. (2024). Faithfulness in Retrieval-Augmented Generation. ACL.

[10] Zhang et al. (2023). Rethinking Layerwise Interventions in LLMs. NeurIPS.
[11] Dai et al. (2024). Controlling Language Models with Activation Engineering. ACL.
[12] He et al. (2026). Detecting Hallucinations in RAG via Semantic-level Internal Reasoning Graph. ArXiv (Jan 2026).
[13] ReDeEP (2025). Detecting Hallucination in RAG via Decoupling External and Parametric Knowledge. ICLR (Jan 2025).
[14] Wang et al. (2025). Attribution-Based Internal Reasoning for RAG Faithfulness. EMNLP (Oct 2025).
[15] Xia et al. (2025). Language Model Uncertainty Quantification with Attention Chain. ACL (Mar 2025).